

Unit-2

Computer Programming Fundamentals

Input Devices

- Input devices are used to take raw data as input from the user.
- Input device enables the user to send data, information, or control signals to a computer. The Central Processing Unit (CPU) of a computer receives the input and processes it to produce the output.

Some of the popular input devices are:

1) Keyboard

The keyboard is a basic input device that is used to enter data into a computer or any other electronic device by pressing keys. It has different sets of keys for letters, numbers, characters, and functions. Keyboards are connected to a computer through USB or a Bluetooth device for wireless communication.

2) Mouse

The mouse is a hand-held input device which is used to move cursor or pointer across the screen. It is designed to be used on a flat surface and generally has left and right button and a scroll wheel between them. Laptop computers come with a touchpad that works as a mouse.

It lets you control the movement of cursor or pointer by moving your finger over the touchpad.

i) Trackball Mouse:



It is stationary input device that has ball mechanism to move cursor on screen. The ball is half inserted in device and can be easily rolled with finger to move pointer on screen. So it is an ideal device if you have limited desk space as you don't need to move it like a mouse.

ii) Cordless or Wireless Mouse:



This type of mouse lacks cable and uses wireless technology such as infrared, Bluetooth or Wi-Fi to control movement of cursor. It is used to improve the experience of using a mouse. It uses batteries for its power supply.

3) Scanner

The scanner uses pictures and pages of text as input. It scans picture or document. The scanned document then converted into a digital format or file and is displayed on screen as an output. It uses optical character recognition techniques to convert images into digital.

i) Flatbed Scanner:



It has a glass pane and a moving optical CIS or CCD array. The light illuminates the pane, and then image is placed on glass pane. The light moves across the glass pane and scans the document and thus produces its digital copy.

ii) Handheld Scanner:



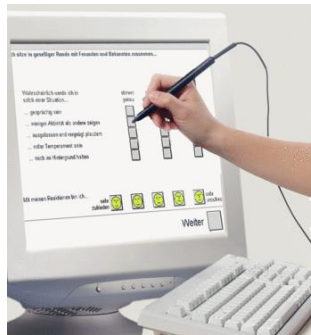
It is a small manual scanning device which is held by hand and is rolled over a flat image that is to be scanned. The drawback in using this device is that hand should be steady while scanning otherwise it may distort image. Example barcode scanner, used in shopping centers.

4) Joystick



It is made up of a stick with a spherical base. The base is fitted in a socket that allows free movement of stick. The movement of stick controls cursor on screen. In joystick, cursor keeps moving in direction of joystick whereas, in mouse, the cursor moves only when mouse moves.

5) Light Pen



A light pen is a computer input device that looks like a pen. The tip of light pen contains a light-sensitive detector that enables the user to point or select objects on display screen. Its light sensitive tip detects the object location and sends corresponding signals to the CPU. It is not compatible with LCD screens, so it is not in use today.

6) Microphone

It is a computer input device that is used to input the sound. It receives sound vibrations and converts them into audio signals or sends to a recording medium. The audio signals are converted into digital data and stored in computer. The microphone also enables the user to

telecommunicate with others. It is also used to add sound to presentations.

i) Dynamic:



It is the most commonly used microphone with a simple design. It has a magnet which is wrapped by a metal coil and a thin sheet on front end of magnet. The sheet transfers vibrations from sound waves to coil and from coil to electric wires which transmit sound like an electrical signal.

ii) Condenser:



It is designed for audio recording and has a very sensitive and flat frequency response. It has a front plate called diaphragm and a back plate parallel to the front plate. When sound hits the diaphragm, it vibrates diaphragm and alters the distance between the two plates. The changes in distance are transmitted as electric signals.

7) Magnetic Ink Character Recognition (MICR)



MICR computer input device is designed to read text printed with magnetic ink. MICR is a character recognition technology that makes use of special magnetized ink which is sensitive to magnetic fields. It is widely used in banks to process cheques and other organizations where security is a major concern.

A document printed in magnetic ink is required to pass through a machine which magnetizes the ink, and the magnetic information is then translated into characters.

10) Touchpad



It is usually found in laptops as a substitute for mouse. It allows you to move or control the cursor on screen using your finger. Just like a mouse, it also has two buttons for right and left click. Using the touchpad, you can perform all the tasks that you do with a mouse.

Output Devices

- The output device displays result of processing of raw data that is entered in the computer through an input device.
- There are number of output devices that display output in different ways such as text, images, hard copies, and audio or video.

1) Monitor

The monitor is display unit or screen of the computer. It is the main output device that displays the processed data or information as text, images, audio or video.

i) CRT Monitor



CRT monitors are based on the cathode ray tubes. Cathode rays tube produces a beam of electrons through electron guns that strike on the inner phosphorescent surface of the screen to produce images on screen. The monitor contains millions of phosphorus dots of red, green and blue color. These dots start to glow when struck by electron beams.

ii) LCD Monitor



The LCD monitor is a flat panel screen that is compact and light-weight as compare to CRT monitors. It is based on liquid crystal display technology which is used in screens of laptops, tablets, smart phones. An LCD screen comprises two layers of polarized glass with a liquid crystal solution between them. When light passes through first layer, electric current aligns liquids crystals. The aligned liquid crystals allow a varying level of light to pass through second layer to create images on screen.

iii) LED monitor



The LED monitor is an improved version of LCD monitor. It also has a flat panel display and uses liquid crystal display technology like LCD monitors. The difference between them lies in the source of light to backlight the display. The LED monitor has many LED panels, and each panel has several LEDs to backlight the display, whereas LCD monitors use cold cathode fluorescent light to

backlight the display. Modern electronic devices such as mobile phones, LED TVs, laptop and computer screens, use LED display as it not only produces greater light intensity but also consumes less power.

2) Projector



A projector is an output device that enables the user to project the output onto a large surface such as a big screen or wall. It can be connected to a computer to project their output onto a screen. It uses light and lenses to produce magnified texts, images, and videos. So, it is an ideal output device to give presentations or to teach a large number of people.

3) Printer

A printer produces hard copies of processed data. It enables the user, to print images, text or any other information onto paper. Printers enable the generation of hard copies for record-keeping, presentations and other purposes, from household to professional settings. Based on the printing mechanism, the printers are of two types: Impact Printers and Non-impact Printers.

- **Impact Printers: They are of two types:**
 - A.Character Printers
 - i. Dot Matrix printers
 - ii. Daisy Wheel printers
 - B.Line printers
 - i. Drum printers
 - ii. Chain printers
- **Non-impact printers: They are of two types:**
 - A.Laser printers
 - B.Inkjet printers

Impact Printer

The impact printer uses a hammer or print head to print character or images onto paper. The hammer or print head strikes or presses an ink ribbon against the paper to print characters and images.

A) Character Printers

Character printer prints a single character at a time or with a single stroke of the print head or hammer. It does not print one line at a time. Today, these printers are not in much use due to their low speed and because only the text can be printed. Character printers are of two types-

i) Dot Matrix Printer



Dot Matrix Printer is an impact printer. The characters and images printed by it are patterns of dots. These patterns are produced by striking ink soaked ribbon against paper with a print head. The print head contains pins that produce a pattern of dots on the paper to form the individual characters. Speed of Dot Matrix printers is around 200-500 characters per second.

ii) Daisy Wheel Printer



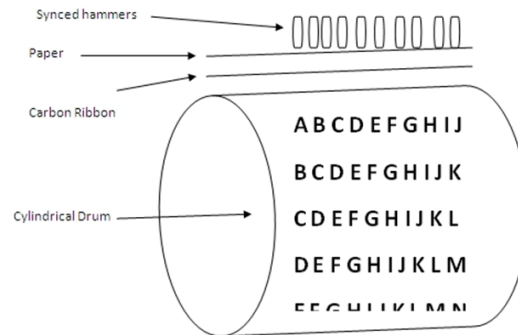
It consists of a wheel or disk that has spokes or extensions and looks like a daisy, so it is named Daisy Wheel printer. At the end of extensions, molded metal characters are mounted. To print a character the printer rotates the wheel, and when the desired character is on the print location the hammer hits disk and the extension hits the ink ribbon against paper to create impression. It cannot be used to print graphics and is often noisy and speed is very low around 25-50 characters per second.

B) Line Printers:

Line printer prints one line at a time. It is a high-speed impact printer as it can print 500 to 3000 lines per

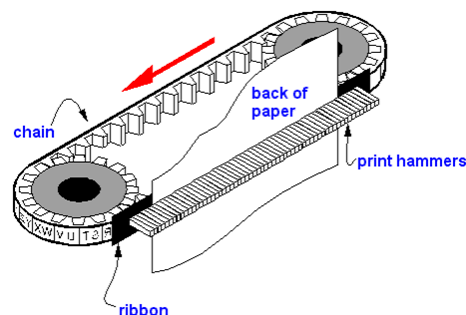
minute. Drum printer and chain printer are examples of line printers.

i) Drum Printer:



Drum printer is a line printer that is made of a rotating drum to print characters. The drum has circular bands of characters on its surface. It has a separate hammer for each band of characters. When you print, drum rotates and when desired character comes under the hammer, hammer strikes ink ribbon against the paper to print characters. The drum rotates at a very high speed and characters are printed by activating the appropriate hammers. Furthermore, it can print only a predefined style as it has a specific set of characters. These printers are very noisy due to the use of hammering techniques.

ii) Chain Printer:



Chain printer is a line printer that uses a rotating chain to print characters. The characters are embossed on the surface of the chain. The chain rotates horizontally around a set of hammers, for each print location one hammer is provided, i.e., the total number of hammers is equal to the total number of print positions.

The chain rotates at a very high speed and when the desired character comes at print location, corresponding hammer strikes the page against ribbon and character on the chain. They can type 500 to 3000 lines per minute. They are also noisy due to the hammering action.

Non-Impact Printer:

Non-impact printers don't print characters or images by striking a print head or hammer on ink ribbon placed against paper. They print characters and images without direct physical contact between the paper and the printing machinery. These printers can print a complete page at a time, so they are also known as page printers.

i) Laser Printer:



A laser printer is a non-impact printer that uses a laser beam to print characters. The laser beam hits the drum,

which is a photoreceptor and draws the image on the drum by altering electrical charges on drum. The drum then rolls in toner and charged image on drum picks the toner. The toner is then printed on the paper using heat and pressure. The laser printers use powdered toner for printing instead of liquid ink and produce quality print objects with a resolution of 600 dots per inch (dpi).

ii) Inkjet Printer:



The inkjet printer is a non-impact printer that prints images and characters by spraying fine, ionized drops of ink. The print head has tiny nozzles to spray ink. The printer head moves back and forth and sprays ionized drops of ink on paper, which is fed through the printer. These drops pass through an electric field that guides ink onto paper to print correct images and characters.

4) Speakers

Speakers are important output devices that transform electrical impulses into audible sounds. The audio system improves multimedia experience by providing high-quality audio output, whether listening to the song, viewing films, playing video games.

Types of speakers

- **Stereo Speakers:** Stereo speakers comprise left and right speakers. They are the most fundamental speaker, found in desktop computers, laptop computers. Stereo speakers offer a simple yet efficient method of reproducing sound.



- **Soundbars:** Soundbars are slim and elongated speakers that provide an all-in-one audio solution for TV/home entertainment systems. Soundbars offer improved audio quality compared to TV speakers.



- **Portable Speakers:** Portable speakers are small, lightweight, wireless devices designed for audio enjoyment. Bluetooth audio systems are available in various styles and sizes with long battery life and waterproof.



5) Headphones/Earphones

Headphones and earphones are personal audio devices worn over the ears or inserted into the ear canal. They deliver Music to the user directly, giving a private and immersive listening experience. Earphones are tiny and fit within the ear, whereas headphones have two speakers that cover both ears. They offer ease, mobility, and ability to concentrate on audio information.

- **Over-Ear Headphones:** Over-ear headphones feature large ear cups that fully envelop the ears. One of their notable advantages is their comfort, as the ear cups completely cover the ears.



- **In-Ear Earphones:** In-ear earphones, known as earbuds or earphones, are inserted right away into ear canal. They are compact, light, transportable. In order to ensure a secure and comfortable fit, in-ear earbuds are available in different sizes.



- **True Wireless Earbuds:** True wireless earbuds are in-ear headphones with no wires connecting them. Each earbud is self-contained and connects to the smartphone through Bluetooth.



6) Plotter

A plotter is a specialized output device used to generate high-quality, accurate, and detailed graphics. It's popular in fields like engineering, architecture, and graphic design. Plotters utilize a pen or marker to draw continuous lines on diverse media such as paper, or film. Plotters are capable of creating accurate drawings, maps, blueprints, and technical diagrams. They are valued for their ability to handle large-format prints and deliver precise output, making them essential tools in industries that require precise graphical representation.

Computer Hardware

Hardware refers to the physical components of a computer or a machine that we can see and touch. It is a physical component that is used in different ways to build a computer or any other machine. The Memory

Devices, Processor, Central Processing Unit, Mouse, and the keyboard all are examples of hardware in computer system. The screen on which you are viewing this page example of hardware. A computer system would not be existing without hardware and not able to run software.

Computer Software

Software is a collection of procedures, instructions, documentation that tells a computer exactly what to do or allows users to interact with a computer. Most of the computers may be useless without software. For example, if a software program, MS-Word is not installed into your computer, you cannot make any document that can be completed through MS-Word. Also, you cannot surf the Internet or visit any website if your system has no Internet browser software. The browser could not run on computer without an operating system. The Google Chrome, MS Word, Excel are examples of software.

Different types of Computer Software

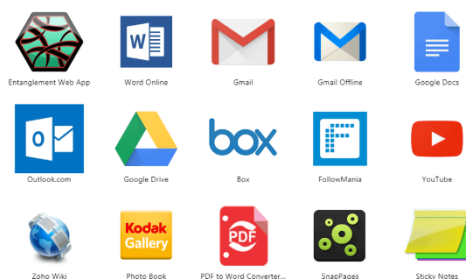
1. System Software



System software is the software that maintains and controls the basic functions of a computer system. It serves as a bridge between the user and the hardware, ensuring the computer runs smoothly. Operating systems, device drivers, and utility programs are a few examples of system software.

- **Operating systems:** An OS, or operating system, is the essential software that controls the resources of computer hardware and applications.
- **Device Drivers:** Device drivers are software elements that help the operating system, and other linked hardware devices communicate with one another.
- **Utility programs:** They are software applications created to perform particular tasks to enhance system performance or offer extra features.

2. Application Software



Application software refers to programs and software applications executing certain tasks or giving end users' features. These programs are made to satisfy users' requirements in various areas, including communication, creativity, entertainment, and productivity.

- **Productivity Software:** project management tools, word processors, spreadsheets, and presentation software programs are all examples of productivity software. Users can create, edit, control files, presentations related activities with those tools.
- **Creative Software:** 3-D modeling programs, photograph- and video-editing programs, image design tools, and music-production software are all examples of creative software.
- **Entertainment Software:** Video games, multimedia players, and streaming programs are examples of entertainment software. These allow users to participate in entertaining, recreational activities.
- **Communication Software:** Email clients, instant messaging programs, video conferencing tools, and web browsers are all examples of communication software, allow users to connect, work together.

The relationship between hardware and software are:

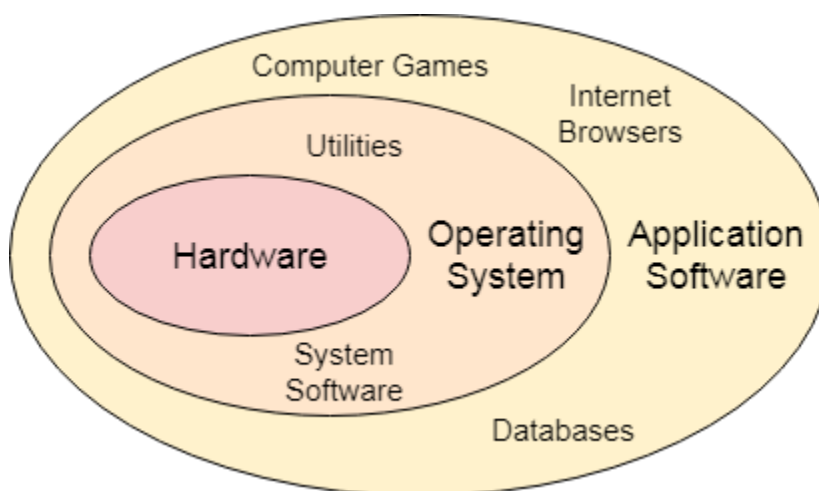
- Hardware and software both are interdependent on each other. Each of them should work along to form computer produce a helpful output.
- The software can not be used if there is no support of any hardware device.
- When there is no proper instructions given, the hardware cannot be used and is useless.
- To get a selected job done on the pc, the relevant software package has to be loaded into the hardware.
- Hardware could be a former expense.
- Software development is incredibly pricey and could be a continued expense.
- The different software package can be loaded on hardware to run totally different jobs.
- The software acts as an associate interface between the user and therefore the hardware.
- We can say the hardware and software are the heart and soul of a digital computer.

Hardware	Software
Hardware is a physical component of computers that executes the instruction.	Software is a program that enables users to interact with the computer, its hardware.
It is manufactured in factories.	It is developed by software programmers or software development companies.
Storage Devices, Input Devices, Output Devices, and Internal components are the primary categories of hardware.	Operating Systems, Application Software, and Programming Software are the main categories of software.
Hardware can be seen and touch as it is a physical, electronic device.	The software can be seen but cannot be touched as it is virtual, not physical.
Computer viruses cannot affect hardware.	Computer viruses can affect software.
Hardware can be replaced with a new one if it is damaged.	The software is reinstalled if it gets damaged.
Through the network, hardware cannot be transferred electrically. Only, it can be physically transferred.	The software can be transferred easily.
Examples of hardware are Keyboard, Printer, Monitor, Mouse, Hard disk and more.	Examples of software are Google Chrome, MySQL, MS Word, Excel, PowerPoint, Notepad, Photoshop and more.

Operating System

- An Operating System can be defined as an interface between user and hardware.
- The operating system is a connecting medium between the user and the hardware.
- It helps the users to access the hardware in an efficient manner.

- It also makes sure that all the computer resources will be used in an efficient and smooth way without any problems.
- For example, when seniors pass their instructions to the juniors, and in the same way, the users gives the instructions to the operating system.
- It is responsible for the execution of all the processes, Resource Allocation, CPU management, File Management and many other tasks.



Functions of Operating System

1.Memory Management

It is the management of the main or primary memory. Whatever program is executed, it has to be present in the main memory. When the program is completed, the memory is released and can be used by other programs. Therefore, there can be more than one program present at a time. Hence, it is required to manage the memory.

The operating system:

- Allocates and deallocates the memory.
- Keeps a record of which part of primary memory is used by whom and how much.
- Distributes the memory while multiprocessing.
- In multiprogramming, operating system selects which processes acquire memory when and how much memory they get.

2.Device Management

Operating system manages the communication between computer system and the peripheral devices connected to I/O port of the system. It also makes sure when a program is executed or terminated, it will stop communication between device and computer.

An operating system regulates device connection using drivers. This management is done by the OS. The OS:

- Allocates and deallocates devices to different processes.
- Keeps records of the devices.
- Decides which process can use which device for how much time.

3.File Management

It specifies which process receives the file and for how long. It also keeps track of information, location, uses, status, and so on. These groupings of resources are

referred to as file systems. The files on a system are stored in different directories. The OS:

- Keeps records of the status and locations of files.
- Allocates and deallocates resources.
- Decides who gets the resources.

4.Process Management

The process is a program under execution. When more than one process runs on the system the OS decides how and when a process will use the CPU. Hence, the name is also **CPU Scheduling**. The OS does

- Allocates and deallocates processor to the processes.
- Keeps record of CPU status.

Certain algorithms used for CPU scheduling are :

- **First Come First Serve (FCFS)**- In this which comes first is served first by the CPU until it is completed.
- **Shortest Job First (SJF)**- process which has shortest execution time will be served first by CPU.
- **Round-Robin Scheduling**- there is a specific time quanta assigned for each process. If process is not executed in time quanta, it is sent to waiting queue for its next turn.
- **Priority-based scheduling**- Processes are given priority based on different criteria and scheduled according to highest priority.

5.Resource Allocation

Operating system manages all resources of computer system. It ensures use of all resources by managing which resource is used by which process and for how much time.

6.Handling the I/O Operations

Input/output devices like mouse, keyboards, printers are external I/O devices and have different natures of each device. The operating system appropriately manages input and output operation of these devices.

7.Scheduling of resources and jobs

Operating system prepares list of tasks to be performed by device of computer system.

- OS decides which device to use for which task. This action becomes complicated when multiple tasks are to be performed in a computer system.
- The scheduling programs of operating system determine order in which tasks are completed.
- It performs these tasks based on priority of performing tasks given by user.

8.Security

Computer security is important aspect of operating system. Reliability of operating system is determined by

how much better security it provides. Modern operating systems use a firewall for security. A firewall is a security system that monitors every activity happening in the computer and blocks that activity in case of any threat.

9. Monitoring activities

Operating system takes care of activities of computer system during various processes. This aborts the program if there are errors. OS sends messages to user for any error in input/output device. It also provides security to system when used in systems operated by multiple users. So that illegal users cannot get data from system.

10. Storage Management

The OS uses storage management for:

- Improving the performance of the data storage resources.
- It optimizes the use of various storage devices.
- Assists businesses in storing more data on existing hardware, speeding up data retrieval process, preventing data loss, and lowering IT costs

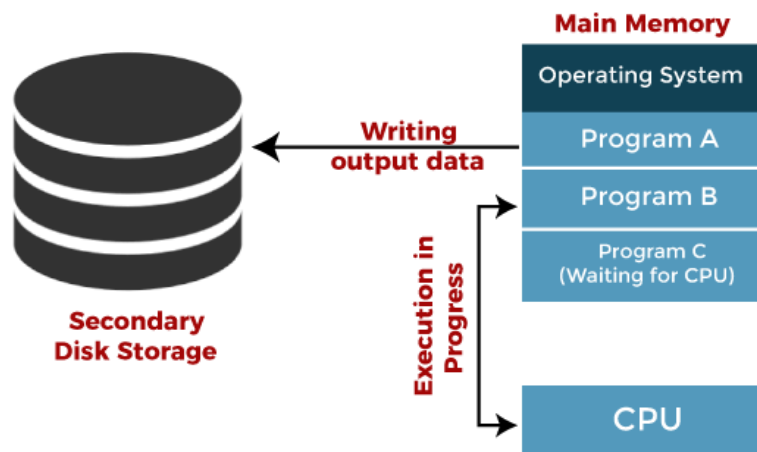
Difference between Single User and Multi-User OS

Terms	Single User Operating System	Multi-User Operating System
Definition	A Single-User Operating System is a system in which only one user can access the computer system at a time.	A Multi-User Operating System is a system that allows more than one user to access a computer system at one time.
Bootloader	The bootloader loads all the resources of the CPU in the profile of a single superuser.	The bootloader distributes the resources of the CPU based on the priority of the users.
Super User	A superuser gets all the powers of maintaining the system and making changes to ensure the system runs smoothly.	Superuser does not exist for a multi-user operating system as each entity has control over their working.
Complexity	Single-User Operating System is simple and easy to design.	The Multi-User Operating systems are complicated as they have to apply complex techniques to share the resources among various users.
Performance	Only one task at one time gets performed.	Schedules different tasks for performance at any rate.
Example	MS-DOS, Windows 95, Windows NT, Windows 2000, Personal Computers, etc.	Mainframes, IBM AS400, Linux & Unix Distributed OS, etc.

Multiprogramming Operating System

- Each process needs two types of system time: CPU time and IO time.
- In multiprogramming, when a process does its I/O, CPU can start execution of other processes.

- Therefore, multiprogramming improves efficiency of the system.



Jobs in multiprogramming system

Advantages of Multiprogramming OS

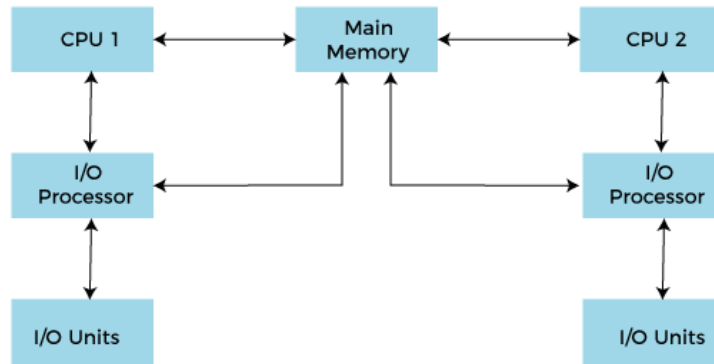
- Overall performance is increased as the CPU always had one program to execute.
- Response time can also be reduced.

Disadvantages of Multiprogramming OS

- Multiprogramming systems provide an environment in which various systems resources are used efficiently, but they do not provide any user interaction with the computer system.

Multiprocessing Operating System

- In Multiprocessing, Parallel computing is achieved.
- There are more than one processors present in system which can execute more than one process at the same time.
- This will increase the throughput of the system.



Working of Multiprocessor System

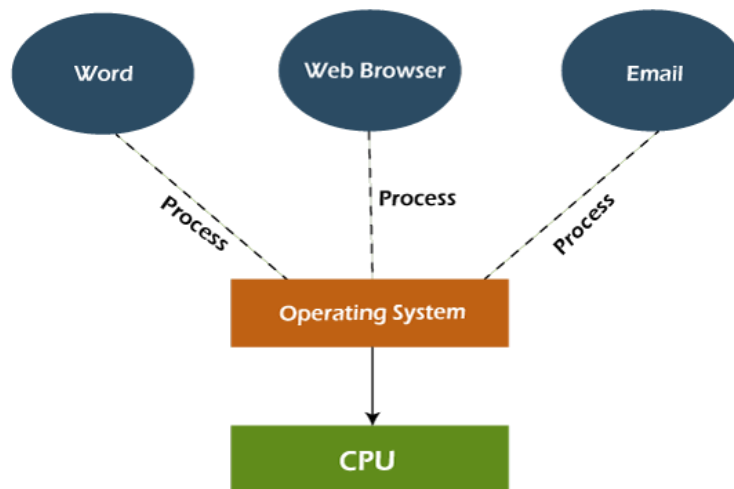
Advantages of Multiprocessing operating system:

- **Increased reliability:** Due to multiprocessing, processing tasks can be distributed among several processors. This increases reliability as if one processor fails, task can be given to another processor for completion.
- **Increased throughput:** As several processors increase, more work can be done in less time.
- **For Example:** UNIX Operating system is one of the most widely used multiprocessing systems.

Disadvantages of Multiprocessing operating System

- Multiprocessing operating system is more complex and sophisticated as it takes care of multiple CPUs simultaneously.

Multitasking Operating System



- The multitasking operating system is a logical extension of multiprogramming system that enables multiple programs to run simultaneously.
- It allows a user to perform more than one computer task at the same time.

Advantages of Multitasking operating system

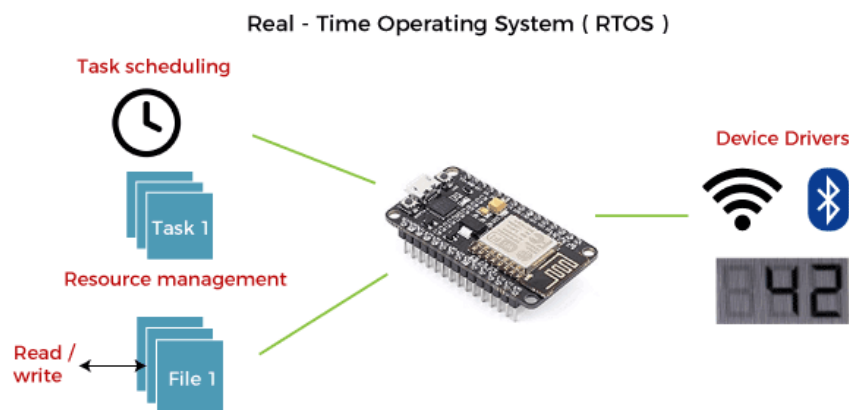
- This operating system is more suited to supporting multiple users simultaneously.
- The multitasking operating systems have well-defined memory management.

Disadvantages of Multitasking operating system

- The multiple processors are busy at the same time to complete any task in a multitasking environment, so the CPU generates more heat.

Real Time Operating System

- In Real-Time Systems, each job carries a certain deadline within which job is supposed to be completed otherwise, huge loss will be there even if the result is produced, it will be completely useless.
- Real-Time system used in military applications, if you want to drop a missile, then missile is supposed to be dropped with a certain precision.



Advantages of Real-time operating system:

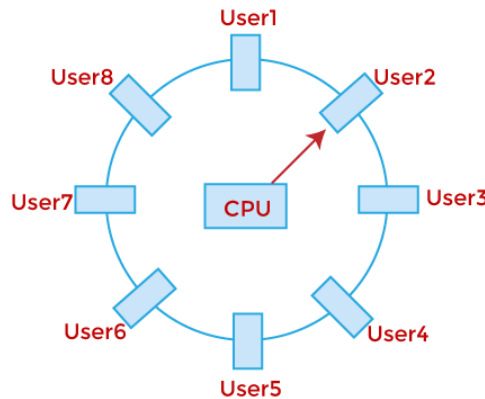
- Easy to layout, develop and execute real-time applications under the real-time operating system.
- In a Real-time operating system, the maximum utilization of devices and systems.

Disadvantages of Real-time operating system:

- Real-time operating systems are very costly.
- Real-time operating systems are very complex and can consume critical CPU cycles.

Time-Sharing Operating System

- In Time Sharing operating system, computer resources are allocated in a time-dependent fashion to several programs simultaneously.
- Thus it helps to provide a large number of user's direct access to the main computer.
- It is a logical extension of multiprogramming.
- In this, CPU is switched among multiple programs given by different users on a scheduled basis.
- A time-sharing operating system allows many users to be served simultaneously, so CPU scheduling schemes , Input/output management are required.



Timesharing in case of 8 users

Advantages of Time Sharing Operating System

- The time-sharing operating system provides effective utilization and sharing of resources.
- This system reduces CPU idle and response time.

Disadvantages of Time Sharing Operating System

- Data transmission rates are very high in comparison to other methods.
- Security and integrity of user programs loaded in memory and data need to be maintained as many users access the system at the same time.
- Time-sharing operating systems are very difficult and expensive to build.

Multithreading Operating System

- A process is a program being executed.
- A process can be further divided into independent units known as threads.
- A thread is like a small light-weight process within a process or collection of threads is called a process.
- Multithreading allows application to divide its task into individual threads. In multi-threads, same process or task can be done by number of threads.

Advantages of Multithreading

- Increased performance and resource utilization.
- With the use of multithreading, multitasking can be achieved.

Disadvantages of Multithreading

- It also introduces complexity and potential issues related to synchronization and concurrency.

Computer Virus

- Computer viruses are unwanted software programs or pieces of code that interfere with the functioning of the computer.
- They spread through contaminated files, data, and insecure networks.
- Once it enters your system, it can replicate to produce copies of itself to spread from one program to another program and from one infected computer to another computer.
- So, we can say that it is a self-replicating computer program that interferes with the functioning of the computer by infecting files, data, programs, etc.

Types of Computer Virus

1. Overwrite Virus

It is the simplest computer virus that overwrites the code of host computer system's file with its own malicious code. The content of infected file is replaced partially or completely without changing the size of file. Thus, it destroys original program code by overwriting it with its defective code. The infected files must be deleted or replaced as this virus cannot be removed or disinfected.

2.Append Virus

This virus appends its malicious code to the end of host program's file. After that, it alters file's header in a way that file's header is redirected to start of the malicious code of the append virus. Thus, this code is executed each time the program runs. However, it does not destroy the host program rather it modifies it in a way that it holds the virus code and enables code to run itself.

3.Macro Virus

Macro virus alters or infects the macros of a document or data file. It is embedded as a macro in a document and adds its codes to the macros of the document. The virus spreads when infected documents are opened in other computers. It also spreads through software programs, which execute macros such as Ms Word, Ms Excel.

4.Boot Virus

Boot virus alters the boot sector program stored in the hard disk or any other storage device such as floppy disks. It replaces the boot sector program with its own malicious version. It infects the computer only when it is used to boot up the computer. If it enters after the boot-up process, it will not infect the computer. Usually, it enters into your system through corrupt media files, infected storage devices and insecure computer

networks. The spread of this virus is very rare these days due to the decline in the use of floppy disk and use of boot-sector safeguards in present-day operating systems.

5.Resident Virus

The resident virus stays permanently in the primary memory (RAM) of computer. When you start the computer, it becomes active and corrupts the files and programs running on the computer.

6.Multipartite Virus

Multipartite virus spreads and infects in multiple ways. It infects both the boot sector and the executable files stored on the hard drive simultaneously. When you turn on a computer, the boot sector virus is triggered as it latches on to the hard drive, which has the data for starting up the computer. Once it is triggered, the program files also get infected.

7.File Infector Virus

It is one of the commonly found computer viruses. It mainly infects the executable files; the files with .com or .exe extensions. The virus becomes active when the infected file is executed. The active virus overwrites the file partially or completely. Thus it may destroy the original file partially or completely.

8.Computer Worm

Computer worm is similar to a virus but is technically different from virus. It can replicate and spread like a virus but unlike viruses, it does not need a host program to spread. Being able to self-replicate it can produce multiple copies of itself. It spreads through networks such as an email sent to an infected email id can infect your system with a computer worm.

9.Trojan Horse

Trojan horse is a malware like a virus or a worm. It can't replicate like virus and worm. Trojan horse hides itself in a program. Once you install any such program, the trojan horse enters into your computer. It can provide unauthorized access to your computer, send your files to other computers, and may delete files or make other unwanted changes in your computer.

10.Cavity virus

It is also known as a space filler virus. As name suggests, this virus tends to install itself by occupying the empty sections of a file. It is not easy to detect this virus as it fills empty spaces without changing the size of the file.

11.CMOS Virus

It infects the CMOS, which stands for complementary metal-oxide semiconductor and is a memory chip that contains the system configuration. This virus can erase or reset the system configuration.

12.Encrypted Virus

It encrypts its payload to make its detection more difficult. It comprises two parts: an encrypted virus body and a decryptor, which decrypts the virus when it is executed. After decryption, the virus can execute itself in order to replicate and become a resident.

13.Executable Virus

It is a non-resident computer virus, which resides in an executable file. Whenever the infected file is executed, it infects the other files.

14.Polymorphic Virus

It creates its thousands of copies itself; in each copy, it changes sequence and byte values to evade detection by antivirus software. Even best antiviruses may not be able to detect this virus. Polymorphic viruses affect data types functions, generally spread through spam infected sites.

15.Rabbit Virus

It is capable of creating new processes, and each new process further creates new processes. This process continues until this virus utilizes all available resources in system and system falls short of resources. It may cause target system to slow down and crash. For eg, it is like an Infinite loop that repeatedly creates processes that consume lots of CPU cycles ,operating system resources.

16.Stealth Virus

It is a hidden computer virus, which specifically attacks operating system processes. It usually hides itself in partitions, files or boot sectors and is capable of going unnoticed during antivirus or anti-malware scans, i.e., it can avoid detection intentionally.

Characteristics of computer viruses

- They can be spread through various means, such as email attachments, infected software downloads, and USB drives.
- They can hide themselves on a computer, making them difficult to detect and remove.
- They can cause harm to a computer's operating system, files, and programs.

- Some viruses are designed to steal personal information, such as login credentials or credit card numbers.
- Some viruses can be used to remotely control an infected computer, part of a botnet.
- Some viruses are designed to spread to as many computers as possible in order to cause widespread damage.
- They are transferred from one element to another.

Antivirus software

- ***Antivirus software*** is a sort of application that defends computers from malware such as viruses, computer worms, trojans, spyware, botnets, rootkits, keyloggers, and other threats.
- Antivirus software scans your computer for viruses, finds them, and removes them.
- Anti-virus software comes in range of versions and formats.
- Antivirus software is designed to protect computers and, once viruses are detected, to destroy them.
- Most of the antivirus software allows for both automated and manual screening.

List of Top 10 Antivirus Software

1. Norton

Each of Norton's antivirus packages provides great protection against malware and system load is significantly reduced. They contain firewall extensive browser protection features to keep you secure online.

2. Bitdefender

It also provides the best value with features such as secure browser with a virtual keyboard, a Wi-Fi network scanner, defense system against encrypting ransomware.

3. Kaspersky

Kaspersky show great malware detection capability and have a quality to moderate effect on the system.

4. McAfee

McAfee Antivirus Plus is only antivirus solution that identifies rising number of devices in a single household.

5. Webroot

Webroot Secure AntiVirus is a fascinating Windows and Mac application that employs unique approach to malware detection.

6. ESET

ESET is a secure antivirus that excels at guarding your gadgets against malware, phishing scams, ransomware and online threats. It also provides excellent capabilities that can assist you in recovering your device if it is lost.

7. Sophos

Sophos is simple antivirus with strong malware detection rates, a user-friendly dashboard, and a few useful features, such as remote management.

8. Avast

Avast offers a variety of features and tools to improve your cybersecurity. Avast's plan includes standard features and functionalities.

9. Panda

Panda Dome is a multi-featured antivirus with a high-quality cybersecurity defenses. It has password organizer which includes password analyzer that examines all of your passwords and informs you how strong they are and how long it will take someone to crack them.

10. BullGuard BullGuard is excellent anti-malware, provides number of handy bonus features and a booster that boosts System speed for gamers.